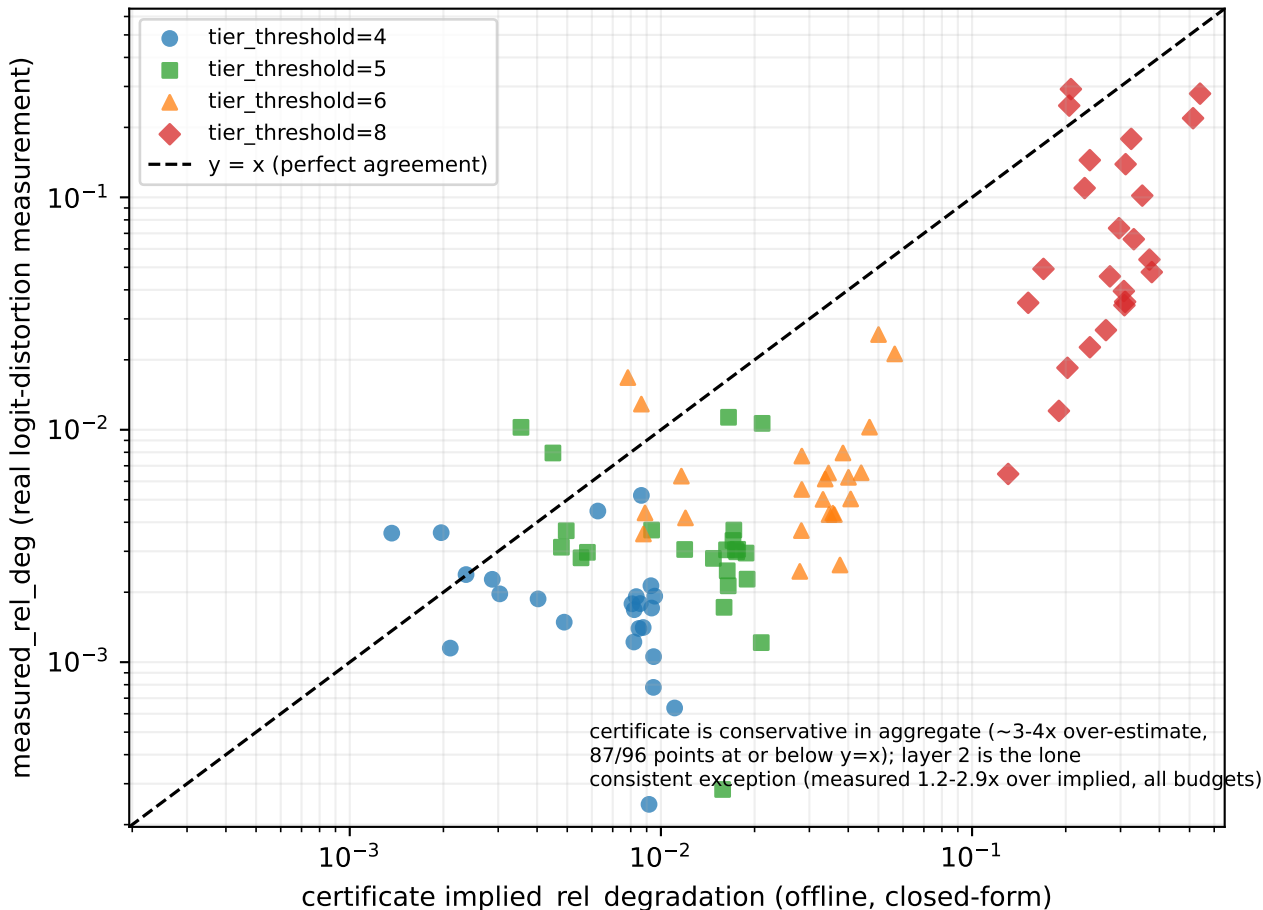


Certificate vs measured (int8-tier degradation, gpt2-scale)



gpt2 scale = mechanism verdict only, not a paper-scale quality claim (Llama fit-side replication pre-registered before any paper claim; see corpus_transfer_verdict.json's gpt2_yellow_flag). results/k4_dec_quant/20260724-125348-0f49e32/cert_vs_measured.parquet, 96 points (87/96 at or below y=x). x = the offline closed-form int8-tier certificate's per-layer implied_rel_degradation (no cache re-scored); y = the same layer's measured relative logit-distortion degradation. One marker per tier_threshold in {4,5,6,8}. The instrument-validation evidence for the cheap-analytic-instruments pattern used throughout K4: the certificate is a conservative over-estimate in aggregate (~3-4x throughout, e.g. T=5 layer-mean implied 1.35%/1.39% vs measured 0.36%/0.43% at b2.2/b2.5, docs/2026-07-25-k4-local-levers-results.md) -- but NOT conservative at every point: layer 2's 8 rows (both budgets, all 4 tier_thresholds) all measure 1.2-2.9x ABOVE their implied value, the one layer where the certificate under-states measured degradation. Ordering (T4<T5<T6<blanket) still holds with zero sign flips at every layer including layer 2 -- the exception is a magnitude miscalibration, not a decision-rule failure. Gating on the aggregate/mean certificate remains conservative; gating on a single layer's certificate value is not guaranteed to be.